Clustering Households by Peak Electricity and Gas Usage Patterns

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Introduction

Household energy consumption varies throughout the day due to daily routines, appliance use, and lifestyle characteristics.

Electricity and gas usage often peak at specific hours reflecting when households are most active. But do these peaks reveal consistent patterns?

This research investigates whether households can be grouped based on their daily peak electricity and gas consumption hours.

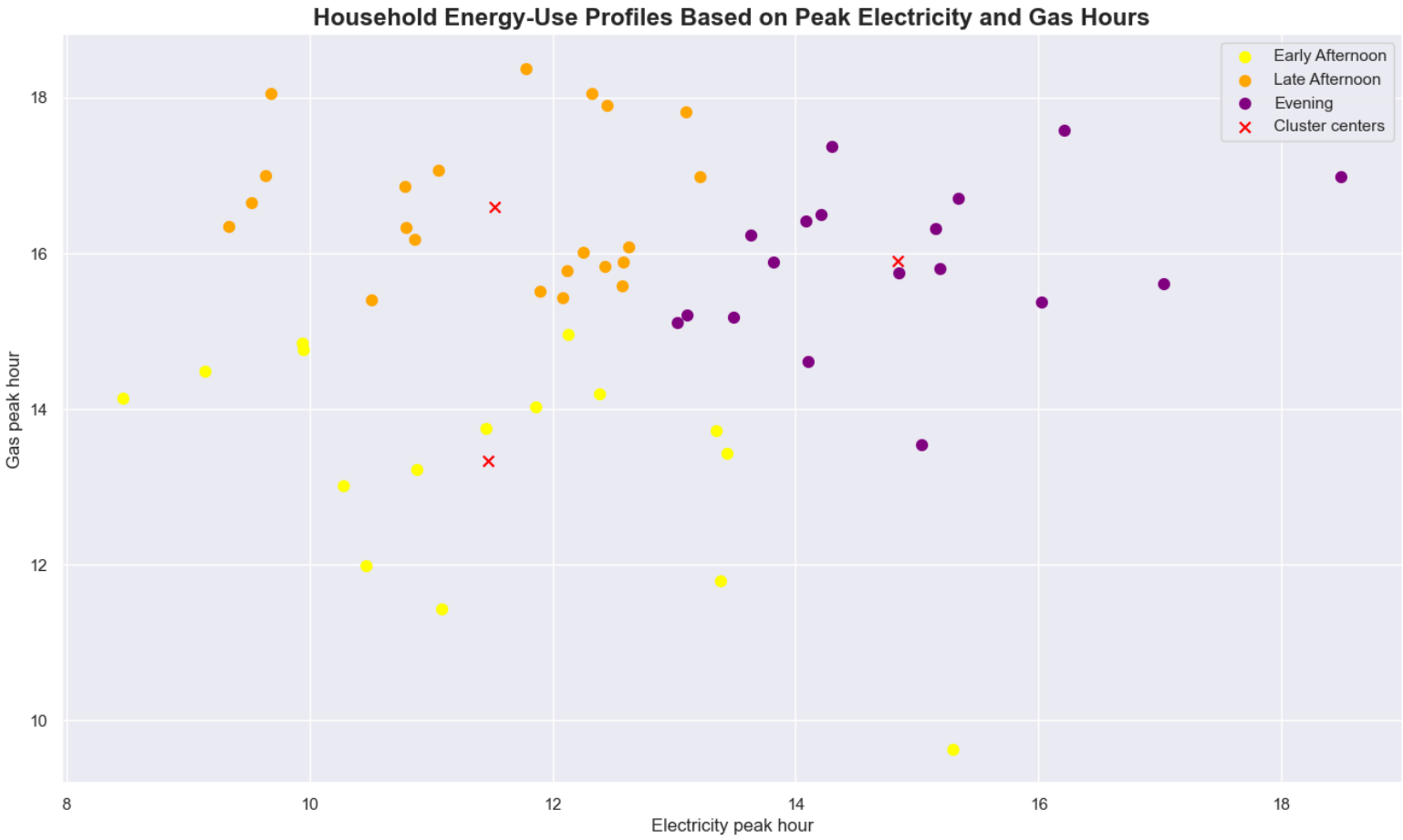
Methods

To answer this question, the k-means clustering algorithm was used to identify natural groupings without needing predefined labels.

Hourly gas and electricity consumption data were first processed to determine each household's daily peak consumption hour for both energy types, and afterwards these daily peak hours were aggregated per household to compute the mean values for each.

Finally, k-means clustering was applied to these aggregated peak hours to group households into distinct energy profiles.

Results



In the scatter plot above, we can observe how the clustering revealed three distinct households' energy-use profiles based on the mean daily peak electricity and gas consumption hours. One group, labeled Early Afternoon is characterized by having early peak hours both in gas and electricity. A second group shows Late Afternoon peaks, while a third group exhibits Evening peak consumption, particularly for electricity. Thus, these clusters indicate that households do differ in the timing of their daily energy use, reflecting their distinct consumption patterns. Overall, the results shows that the data is sufficient to meaningfully group households.

Discussion

Although this approach takes advantage of daily peak consumption hours as a simple and interpretable way to summarize household behavior to find well suited distinct energy-use profiles. It fails to capture the magnitude of energy consumption, meaning households with similar peak timing but very different usage levels may be grouped together. To improve the analysis and avoid bias on weekday data points we could separate the analysis to capture differences in routines and further clustering validation methods could also strengthen the robustness of the identified profiles.

Conclusion

Overall, from the results and identified clusters it can be concluded that there does exist meaningful groupings based on daily peak electricity and gas consumption hours. This works to show how households tend to concentrate their energy use within distinct time periods, such as early afternoons, late afternoons, or evenings. Therefore, indicating that households exhibit distinct timing-based energy-use profiles.